CS 4243/5493 Large Scale Database Management Systems

Project: A Database Application with a Web Front-end
Part 1: Design Activities and QA

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Part 1 Database Design

What database application? Based on interests and availability of data

Requirements: Entity sets (>=8), relationship sets (>= 8), business rules

Where can I find data to populate my database? Real-world data from Kaggle or other platforms, or use AI-assisted generation of synthetic data

How will users interact with database? Describe it in the business rules

Prefer **Relational DBMS** (PostgreSQL), Non-SQL applications are allowed but require special approval.

ML or LLMs are optional to add and recommend if predictive modeling is essential over the data.

Part 1 Design Submission

Submit a ZIP file including

- ✴ docx/doc (no more than 2 pages) for database application description
- ✴ ER diagram (scanned version of hand drawing, or created by online tools)
- ✴ schema.sql

Part 1 due on Oct 21 (Wednesday), **no late submissions**

Submit for feedbacks and adjustment before Part 2

Part 2 Database Application Implementation

- Incorporate feedback, *make sure description is consistent with ER*
- Implement a data loading procedure, e.g., *insertion.sql,* enough tuples (>=100)
- Think about SQL queries
- Design user interactions (recommending python stream lit library)

Requirements:

➞ >= 5 kinds of interactions

➞ >= 5 queries involving a join, a group by, or both.

Part 2 Database Application Submission

Project presentation on Dec 3 (Thursday), ~5 mins

Part 2 due on Dec 3 (Thursday), the same day as the last presentations, **no late submissions**

Submit a ZIP file including

- final project report (no more than 4 pages) in PDF
- data (if accessible)
- code (python files, insertion.sql, README)
- presentation slides in PDF

Presentation Guidelines

Presentation date: **Dec 1 & 3 (Tuesday, Thursday), in person in the classroom**

Schedule Presentation

Duration: 5 mins for each project

Material: ~5 slides to describe your project

- ✴ target application
- ✴ design of the database (i.e., ER diagram and relational tables)
- ✴ loaded data (simulated or from real-world collection)
- ✴ user interface and interaction (e.g., supported queries)

A live demo of your application is encouraged (but not required)

Extra credits: the most favorite (3 extra credits) and second favorite (2 extra credits) projects